Cocaine Usage by Demographics

By: Stephen Lawrence, Charlie Day, Terry Yi, and Rachael Gu

Introduction

- In 2012, the age-adjusted rate of drug overdose deaths involving cocaine was 1.2 per 100,000. In 2019 it was 4.9. 2020, 6.0.
- Target: state-level health department policymaker
- Well separated clustering, fuzzy clustering both useful

Dataset Discussion

- 867 observations, 50 states + D.o.C * 17 years of observations
- Rows represent an individual state per for a given year, data on alcohol, tobacco, marijuana, and cocaine use given
- Data from NSDUH study
- Dataset excluded other drugs collected in the study
- We will only use cocaine and population variables with categorical identifiers (year, state)

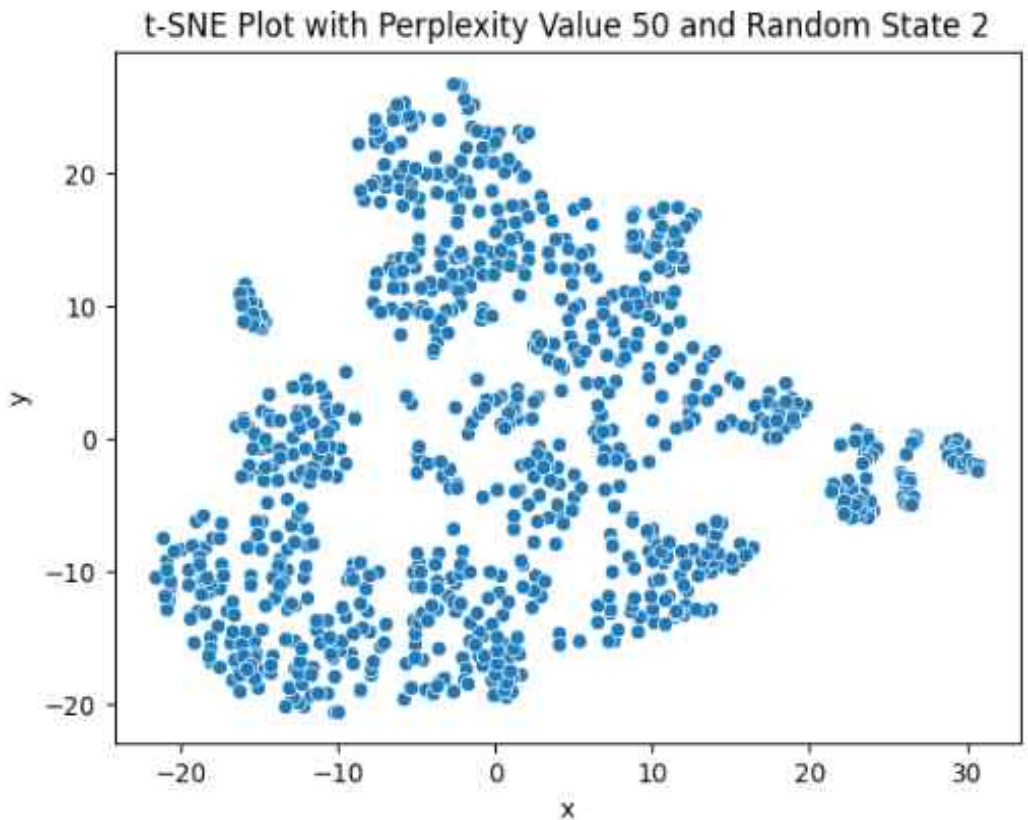
Cleaning and Scaling

- No missing values, no need to reduce
- Variables have drastically different magnitudes, so we should scale

	Population.12-17	Population.18-25	Population.26+	Totals.Illicit Drugs.Cocaine Used Past Year.12-17	Totals.Illicit Drugs.Cocaine Used Past Year.18-25	Totals.Illicit Drugs.Cocaine Used Past Year.26+	Rates.Illicit Drugs.Cocaine Used Past Year.12-17	Rates.Illicit Drugs.Cocaine Used Past Year.18-25	Rates.Illicit Drugs.Cocaine Used Past Year.26+
count	8.670000e+02	8.670000e+02	8.670000e+02	867.000000	867.000000	867.000000	867.000000	867.000000	867.000000
mean	4.897141e+05	6.588800e+05	3.874155e+06	5.062284	37.107266	59.107266	10.280396	56.923988	14.968755
std	5.637959e+05	7.559898e+05	4.320776e+06	7.512625	46.663047	72.738872	5.499766	18.145774	5.256846
min	3.055100e+04	5.739500e+04	3.101100e+05	0.000000	2.000000	2.000000	1.838000	18.345000	5.303000
25%	1.315405e+05	1.742935e+05	1.027871e+06	1.000000	10.000000	14.000000	5.638000	42.849000	11.534000
50%	3.396850e+05	4.562400e+05	2.698757e+06	3.000000	22.000000	36.000000	8.820000	55.660000	14.468000
75%	5.410950e+05	7.468080e+05	4.509094e+06	6.000000	46.000000	75.000000	14.448000	67.479500	17.212000
max	3.293484e+06	4.469106e+06	2.591772e+07	56.000000	345.000000	585.000000	32.411000	122.375000	52.488000

Clusterability

- High variability across states and demographics
- Significant correlations between age group populations and cocaine usage rates
- State specific data over multiple years boosts clustering potential.
- Diverse variables, like demographics and drug usage rates, aid in nuanced cluster identification
- Standardized scaling ensures equal variable contribution, enhancing the effectiveness of distance-based clustering

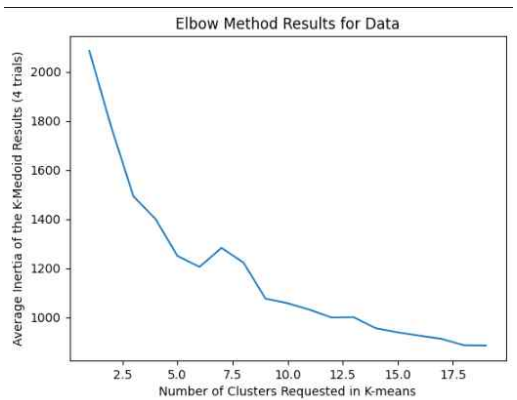


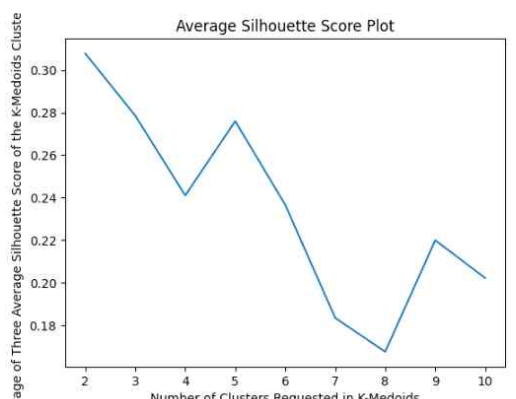
Algorithm Selection

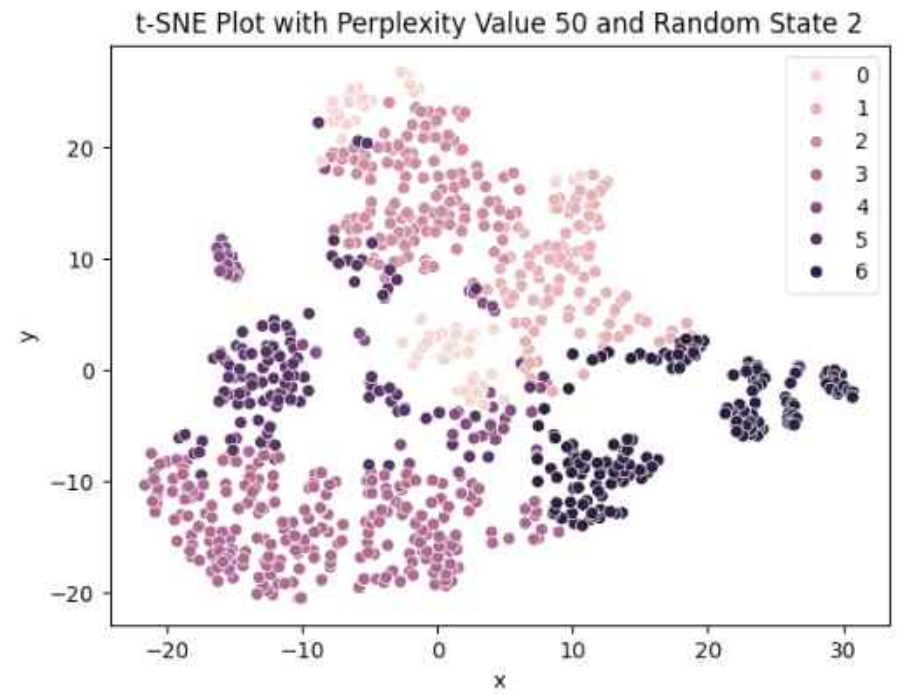
1. K-Medoids
. Robust to Outliers
. Non-Spherical Clusters
. Stability

1. Fuzzy clustering
. Good fit for analyzing drug usage patterns
. Capture complex and overlapping influences

K-Medoids


Elbow Method Results for Data
Average Inertia of the K-Medoid Results (4 trials)
2000
1800
1600
1400
1200
1000
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Number of Clusters Requested in K-means


Average Silhouette Score Plot
0.30
0.28
0.26
0.24
0.22
0.20
0.18
2
3
4
5
6
7
8
9
10
Number of Clusters Requested in K-Medoids


t-SNE Plot with Perplexity Value 50 and Random State 2
0
1
2
3
4
5
6
y
20
10
0
−10
−20
x
−20
−10
0
10
20
30

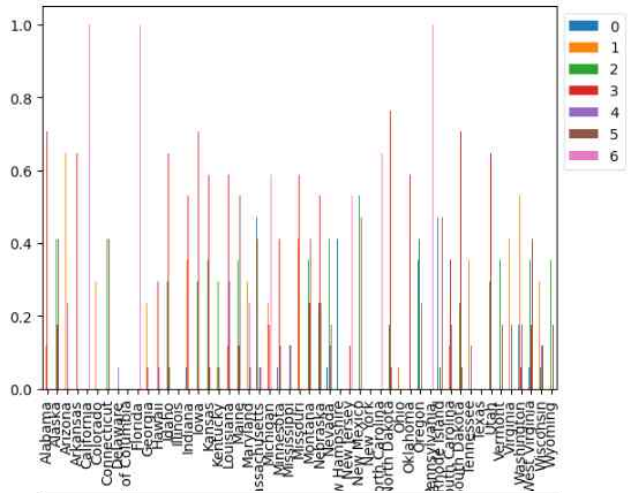
Analysis Result

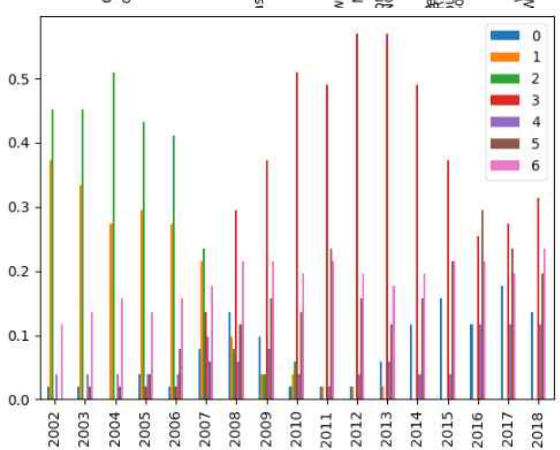
Key observation

- Dominant Cluster 6(t-SNE, similarity matrix)
- Rates of Cocaine Usage
  - Age 12-17
  - Age 18-25

Cluster Attributes

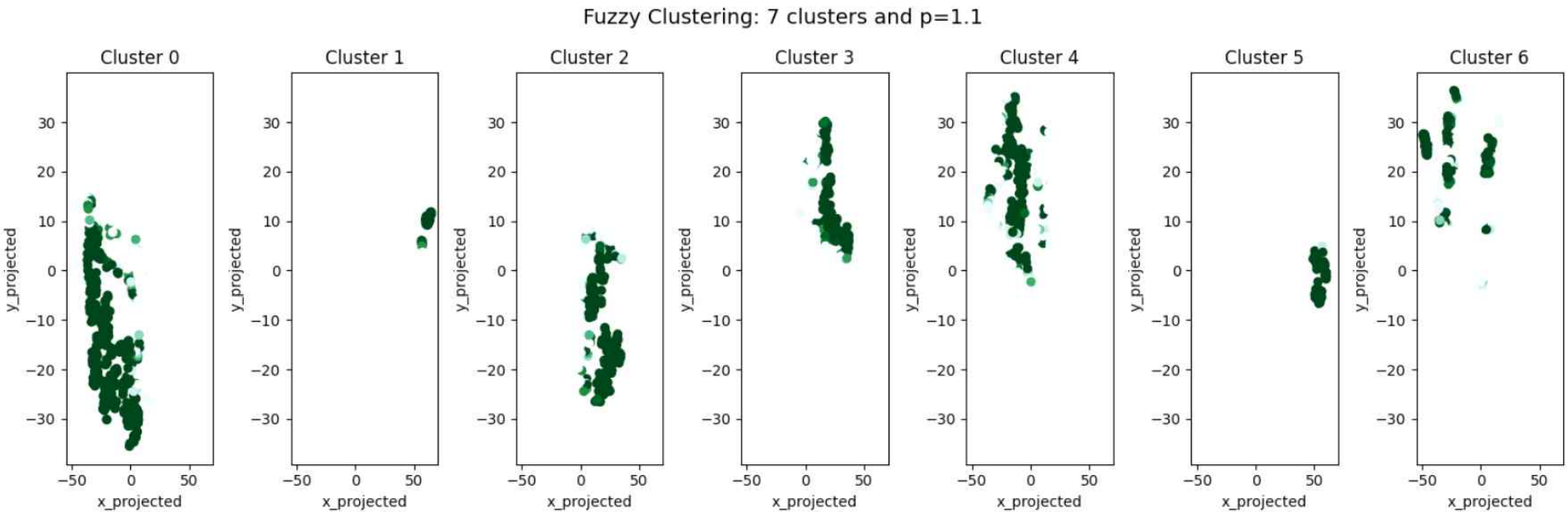
- Cluster 0: Year (2010-2018), diverse array of states(California, Texas), regions with persistent cocaine usage.
- Cluster 1: Year (2003-2018), populous states (New York, Florida), consistent cocaine usage across extended period.
- Cluster 2: Year (2008-2018), lower population densities, emerging reports of cocaine usage.
- Cluster 3: Year (2002-2007), successful interventions in drug trends post-2007(Ohio ,Pennsylvania) .
- Cluster 4: Year (2002-2007), earlier peaks in cocaine usage trends(Oregon, Nevada).
- Cluster 5: Year (2003-2018), consistent cocaine usage issues without significant fluctuations(Michigan, Illinois).
- Cluster 6: Year (2000-2018), consistent cocaine usage of lower population states(Montana, Wyoming).





Fuzzy c-means

내용을 입력하십시오



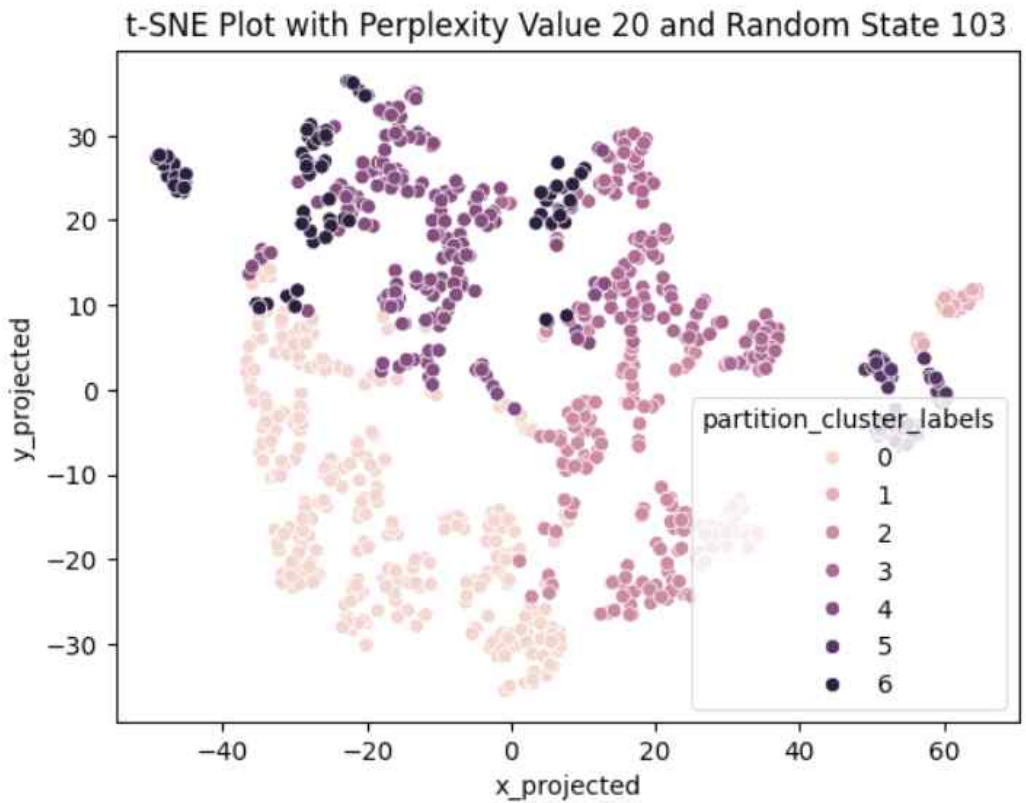
Partitional Clustering

Key Observations:

- don't follow ideal conditions: size, shape, separation, etc.
- some clusters have been split up (e.g. 6)

Cluster Attributes:

- Cluster 0: Year (2008 - 2018) and total illicit cocaine usage
- Cluster 1: high population among all age categories
- Cluster 2: Year (2008 - 2018), relative low population and rate of usage
- Cluster 3: Year (2002 - 2007), relative low population and rate of usage
- Cluster 4: Year (2002 - 2007), low population and rate of usage
- Cluster 5: moderate population and total cocaine usage
- Cluster 6: low population, low total usage, and moderate rate of cocaine usage



Discussion

- Rand-score indicates different results, further analysis suggests otherwise
- Due to similarity between clustering and high dimensionality, both are not fit
- K-Medoids well separated, more trust
- K-Medoids revealed insights on geographic locations
- Fuzzy c-means revealed points being clustered in multiple groups

Conclusion

Observations:

- No inherent clusters identified

Shortcomings:

- ‘Population’ had a large magnitude -> biased initial clustering results
- same state and close years are similar

Future Work:

- Restructure data
- Analyze cocaine data with synthetic opioid data

Thank You!